

How much does the endpoint say about the switching rate?

Working note for direction S1 (moved out of Paper C, October 2026)

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1 Setting

This note was Section 8 and Appendix E of an earlier version of Paper C (“Face selection for slowly switching Markov chains and the planar persistent walk”). The mathematics is unchanged. It uses only Paper A [2], and this section recalls what it needs from there. Smiga, Radaelli, Binder and Landi [4] compute the Fisher information of the empirical distribution under a Gaussian approximation. Their leading term vanishes for the symmetric walk of Paper A, whose stationary law does not depend on the switching rate.

Kagey’s board [1] has N steps. The first step is left or right with probability $\frac{1}{2}$, and each later step repeats the previous one with probability p . We write $q = 1 - p$ and $u = q/p$, the odds of a switch. We write $L = \log N$, and $T = Nq$ is the expected number of switches (up to the first step). $P_N(k)$ is the probability of k right steps, that is, that the ball lands in bin k . The endpoints have $P_N(0) = P_N(N) = p^{N-1}/2$.

For $a, b \geq 1$ let $S_{a,b}(u) = \sum_w u^{s(w)}$, the sum over the words w with a right steps and b left steps, where $s(w)$ is the number of switches of w . Then $S_{a,b}(q/p)$ is the ratio $P_N(a)/P_N(0)$ of the probabilities of bin a and bin 0 after $a + b = N$ steps (the run count of [2]). Let $S_N = S_{\lfloor N/2 \rfloor, \lceil N/2 \rceil}$. It has nonnegative coefficients and no constant term, so $S_N(u) = 1$ has exactly one positive root u_N , and $p_N = 1/(1 + u_N)$ is the crossing of the middle and the endpoints. Put $z_N = N(1 - p_N)$ and $c_0 = \frac{1}{2} \log(\pi/8)$. Paper A proves that $z_N = L - \frac{1}{2} \log L + c_0 + O(\log L/L)$. We call this the second-order expansion of z_N .

2 The information in the endpoint

This note asks a statistical question about Kagey’s board. If we only see the bin where the ball lands, how much do we learn about the switching probability? The answer is that the bin keeps about a fraction $1/(2T)$ of the Fisher information of the whole path, so about $1/(2 \log N)$ at Kagey’s crossing. We use the notation of Section 1. The first step is left or right with probability $\frac{1}{2}$ each, and each later step repeats the one before it with probability $p = 1 - q$, where $0 < q < 1$, and $T = Nq$. Put $\rho = 1 - 2q$, $u = q/p$ and $y = Nu = T/p$. Let J be the number of switches, K the number of right steps, $X = 2K - N$ the endpoint and $W = X/N$. A path with J switches has probability $\frac{1}{2} p^{N-1-J} q^J$. So $J \sim \text{Bin}(N-1, q)$, and the path carries the Fisher information $(N-1)/(pq)$ about q . The endpoint carries $I_K(q) = \sum_{k=0}^N (\partial_q \mathbb{P}(K=k))^2 / \mathbb{P}(K=k)$, and we call $e_N(q) = I_K(q) pq / (N-1)$ the *efficiency* of the endpoint. It is the fraction of the information left when we see only the endpoint.

The endpoint sees the switches only through how far the walk spreads. A normal X with mean 0 and variance Np/q , close to $\text{Var } X$ for large T , would carry the information $\frac{1}{2}(\partial_q \log(Np/q))^2 = 1/(2p^2q^2)$, which is the fraction $1/(2(N-1)pq) \approx 1/(2T)$. Theorem 2.3 confirms this.

Proposition 2.1 (a correlation ratio). *For $N \geq 2$ and $0 < q < 1$,*

$$e_N(q) = \frac{\text{Var}(\mathbb{E}[J | K])}{\text{Var } J}.$$

For $0 < k < N$ we have $\mathbb{E}[J | K = k] = \frac{d}{dt} \log S_{k,N-k}(e^t)$ at $t = \log(q/p)$, where $S_{a,b}$ is the polynomial of Section 1. For $k \in \{0, N\}$ we have $J = 0$.

Proof. The score of a path is $(J - (N-1)q)/(pq)$. Since $\mathbb{P}(K = k)$ is a sum of path probabilities, $\partial_q \log \mathbb{P}(K = k) = \mathbb{E}[(J - (N-1)q)/(pq) | K = k]$. As $\mathbb{E}J = (N-1)q$, this gives $I_K(q) = \text{Var}(\mathbb{E}[J | K])/(pq)^2$, and we divide by $(N-1)/(pq) = \text{Var } J/(pq)^2$. By the run count of [2], for $0 < k < N$ the probability $\mathbb{P}(K = k, J = j)$ is $\frac{1}{2}p^{N-1}u^j$ times the number of words with k right steps, $N-k$ left steps and j switches. Summing over j gives $\mathbb{P}(K = k) = \frac{1}{2}p^{N-1}S_{k,N-k}(u)$, so $\mathbb{E}[J | K = k] = uS'_{k,N-k}(u)/S_{k,N-k}(u)$. Only the two constant paths have $K \in \{0, N\}$. $\square$

This identity, checked in Lean, says that e_N is the correlation ratio of J on K . Since $\mathbb{E}[J | K]$ is the orthogonal projection of J onto the functions of K in L^2 , the Cauchy–Schwarz inequality gives

$$e_N(q) \geq \frac{\text{Cov}(J, g(K))^2}{\text{Var } J \text{Var } g(K)} \quad \text{whenever } \text{Var } g(K) > 0. \quad (1)$$

Proposition 2.2 (few switches). *(a) For fixed $N \geq 3$, $e_N(q) = 1 - \frac{N-2}{2}q + O_N(q^2)$ as $q \rightarrow 0$.*

(b) Let $q \rightarrow 0$ with $T = Nq$ bounded. Then $e_N(q) \rightarrow 1$ if and only if $T \rightarrow 0$.

In words, the endpoint loses information as soon as paths with two switches appear. The reason is that one switch and two switches both leave the endpoint uniform on the interior bins, so an interior bin cannot tell them apart.

Proof. By Proposition 2.1, $1 - e_N = \mathbb{E} \text{Var}(J | K) / \text{Var } J$, and $\text{Var } J = (N-1)pq$. Fix an interior bin $0 < k < N$. Exactly two words with k right steps have one switch, and exactly $N-2$ have two switches ($k-1$ that start to the right and $N-k-1$ that start to the left). So

$$\mathbb{P}(J = 1, K = k) = p^{N-2}q, \quad \mathbb{P}(J = 2, K = k) = \frac{N-2}{2}p^{N-3}q^2.$$

(a) Here N is fixed. Given $K = k$ the number of switches is 1, or 2 with probability $\frac{N-2}{2}q + O_N(q^2)$, or larger with probability $O_N(q^2)$. So $\text{Var}(J | K = k) = \frac{N-2}{2}q + O_N(q^2)$, and $\mathbb{P}(0 < K < N) = (N-1)q + O_N(q^2)$. For $K \in \{0, N\}$ the conditional variance is 0. Hence $\mathbb{E} \text{Var}(J | K) = \frac{(N-1)(N-2)}{2}q^2 + O_N(q^3)$, and dividing by $(N-1)pq$ gives the claim.

(b) For the upper bound we replace $\mathbb{E}[J | K]$ by the indicator of an interior bin, which can only raise the mean squared error. This gives

$$\mathbb{E} \text{Var}(J | K) \leq \mathbb{E}(J - \mathbf{1}\{0 < K < N\})^2 = \mathbb{E}[(J-1)^2; J \geq 2] \leq \mathbb{E}[J(J-1)] = (N-1)(N-2)q^2,$$

because $J \geq 1$ exactly on the interior bins. So $1 - e_N \leq (N-2)q/p \leq T/p$, which tends to 0 if $T \rightarrow 0$.

For the lower bound, a variable that takes two values 1 apart with probabilities a and b has variance at least ab . So $\text{Var}(J \mid K = k) \geq \mathbb{P}(J = 1 \mid K = k)\mathbb{P}(J = 2 \mid K = k)$, and by the Cauchy-Schwarz inequality

$$\mathbb{E} \text{Var}(J \mid K) \geq \sum_{0 < k < N} \frac{\mathbb{P}(J = 1, K = k)\mathbb{P}(J = 2, K = k)}{\mathbb{P}(K = k)} \geq \left(\sum_{0 < k < N} \sqrt{\mathbb{P}(J = 1, K = k)\mathbb{P}(J = 2, K = k)} \right)^2 = \frac{(N - 1)pq}{2}.$$

Dividing by $(N - 1)pq$ gives $1 - e_N \geq \frac{(N-1)(N-2)}{2} p^{2N-6} q^2$. If T stays in an interval $[t_0, t_1]$ with $t_0 > 0$, the right side tends to a limit between $\frac{1}{2}t_0^2 e^{-2t_1}$ and $\frac{1}{2}t_1^2$, and it is positive. So e_N does not tend to 1 along any sequence on which T stays away from 0. $\square$

Theorem 2.3 (many switches). *Let $R_N^2(q) = \text{Cov}(J, X^2)^2 / (\text{Var } J \text{ Var } X^2)$ be the squared correlation of J and X^2 . For $N \geq 2$ and $0 < q < 1$,*

$$R_N^2(q) = \frac{1}{2(N-1)pq} \cdot \frac{[1 + \rho^N - p(1 - \rho^N)/T]^2}{V},$$

where

$$V = 1 - \frac{1 + 8\rho + \rho^2 + 8\rho^{N+1}}{4pT} + \frac{\rho(1 - \rho^N)[2(1 + 2\rho)(2 + \rho) - \rho(1 - \rho^N)]}{8p^2T^2}.$$

There are absolute constants C and N_0 such that for $N \geq N_0$, $0 < q \leq \frac{1}{2}$ and $1 \leq T \leq N^{1/4}$,

$$0 \leq e_N(q) - R_N^2(q) \leq C \left(\frac{1}{T^3} + \frac{T^3}{N} \right).$$

Consequently, in the same range and with an absolute implied constant,

$$e_N(q) = \frac{1}{2T} + \frac{1}{4T^2} + O\left(\frac{1}{T^3} + \frac{T^3}{N}\right).$$

In words, when T is large but at most $N^{1/4}$, the endpoint keeps about the fraction $1/(2T)$ of the information, and almost all of what it keeps is carried by the single statistic X^2 . The lower bound $e_N \geq R_N^2$ is (1) with $g(K) = X^2$, for all $N \geq 2$ and $0 < q < 1$. For the upper bound, Proposition 2.1 and the uniform Bessel estimate for every bin in [2] (recalled in (2) below) make $\mathbb{E}[J \mid K]$ close to $y\sqrt{1 - W^2} + \frac{1}{2}$. As W is of order $T^{-1/2}$, this is close to $y(1 - W^2/2) + \frac{1}{2}$, which lies in the span of 1 and X^2 . Appendix A proves the closed form and the bound.

Proof of the expansion in Theorem 2.3. If $T \leq 18$, then $|e_N - \frac{1}{2T} - \frac{1}{4T^2}| \leq 1 \leq 18^3 T^{-3}$, since e_N and $\frac{1}{2T} + \frac{1}{4T^2}$ lie in $[0, 1]$. Let $T \geq 18$. As $q \leq \frac{1}{2}$, we have $p \geq \frac{1}{2}$ and $0 \leq \rho < 1$, and $\rho^N \leq e^{-2Nq} = e^{-2T} \leq T^{-2}$. The middle term of V lies in $[0, 9/T]$, and since $1 + 8\rho + \rho^2 = 10 - 20q + 4q^2$ it equals $\frac{5}{2T} + O(q/T + T^{-2})$. The last term lies in $[0, 9/T^2]$. So $V \geq \frac{1}{2}$, $1/V = 1 + \frac{5}{2T} + O(q/T + T^{-2})$, the squared bracket is $1 - \frac{2}{T} + O(q/T + T^{-2})$, and $1/(2(N-1)pq) = (1 + O(q + 1/N))/(2T)$. Multiplying, $R_N^2 = \frac{1}{2T} + \frac{1}{4T^2} + O(T^{-3} + q/T + 1/(NT))$ with absolute constants. Since $q/T = 1/N \leq T^3/N$, the bound on $e_N - R_N^2$ gives the expansion. $\square$

Corollary 2.4 (at Kagey's crossing). *Let p_N be the value of p at which the middle bin and the endpoints are equally likely, and let $q_N = 1 - p_N$ and $z_N = Nq_N$ (Section 1). Then*

$$e_N(q_N) = \frac{1}{2L} + \frac{\log L + 1 + \log(8/\pi)}{4L^2} + O\left(\frac{(\log L)^2}{L^3}\right).$$

Proof. Since $S_N(u) \geq 2u$, we have $u_N \leq \frac{1}{2}$, so $p_N \geq \frac{2}{3}$ and $q_N \leq \frac{1}{3}$. By the second-order expansion of z_N in [2], $z_N = L(1-\varepsilon)$ with $\varepsilon = (\frac{1}{2} \log L - c_0)/L + O(\log L/L^2)$. So Theorem 2.3 applies with $T = z_N$ for large N and gives $e_N(q_N) = \frac{1}{2z_N} + \frac{1}{4z_N^2} + O(L^{-3})$. Here $\frac{1}{2z_N} = \frac{1}{2L} + \frac{\log L - 2c_0}{4L^2} + O((\log L)^2/L^3)$, $\frac{1}{4z_N^2} = \frac{1}{4L^2} + O(\log L/L^3)$ and $-2c_0 = \log(8/\pi)$. $\square$

So $e_N \sim 1/(2 \log N)$ there, but the relative correction is of order $(\log \log N)/\log N$ and is large at every N we computed. The product $2Le_N$ is 1.74 at $N = 100$ and 1.35 at $N = 3200$ (Table 1).

N	z_N	$e_N(q_N)$	$2z_N e_N$	$2Le_N$	$R_N^2(q_N)$	$\frac{1}{2z_N} + \frac{1}{4z_N^2}$
50	2.8998	0.24117	1.3987	1.8869	0.20873	0.20216
100	3.5034	0.18869	1.3221	1.7379	0.16642	0.16309
400	4.7501	0.12681	1.2047	1.5196	0.11715	0.11634
1600	6.0246	0.09433	1.1366	1.3919	0.09007	0.08988
3200	6.6684	0.08361	1.1151	1.3496	0.08070	0.08060

Table 1: The efficiency of the endpoint at Kagey's crossing $q_N = 1 - p_N$, computed from the exact law, with $z_N = Nq_N$ and $L = \log N$. The last two columns are the squared correlation of Theorem 2.3 and its expansion to two terms.

Many balls

Now drop n independent balls and record only the histogram of their bins. How well can q be estimated? By the Cramér–Rao bound no unbiased estimator has variance below $1/(nI_K(q))$, and the maximum likelihood estimator attains this as $n \rightarrow \infty$. The next corollary says that a much simpler estimator does almost as well. Put $m(q) = \mathbb{E}_q X^2$, a polynomial in q .

Corollary 2.5 (the moment estimator). *Let $X_1, \dots, X_n$ be the endpoints of n independent balls with switching probability q , and suppose that $R_N^2(q) > 0$. Then $m'(q) \neq 0$, and with probability tending to 1 the equation $m(\hat{q}) = \frac{1}{n} \sum_{i=1}^n X_i^2$ has a solution $\hat{q}_n$ that converges to q in probability. For this solution*

$$\sqrt{n}(\hat{q}_n - q) \Rightarrow N\left(0, \frac{pq}{(N-1)R_N^2(q)}\right) \quad (n \rightarrow \infty).$$

So its asymptotic efficiency, relative to the bound $1/(nI_K(q))$, is $R_N^2(q)/e_N(q)$. There is an absolute constant C' such that, with N_0 as in Theorem 2.3, for $N \geq N_0$, $0 < q \leq \frac{1}{2}$ and $1 \leq T \leq N^{1/4}$,

$$1 - C' \left(\frac{1}{T^2} + \frac{T^4}{N} \right) \leq \frac{R_N^2(q)}{e_N(q)} \leq 1.$$

In words, among all estimators that see only the bins, matching the mean square of the endpoint loses a fraction of order $1/T^2 + T^4/N$ of the information. At Kagey's crossing the efficiency R_N^2/e_N is 0.87 at $N = 50$ and 0.97 at $N = 3200$ (Table 1).

Proof. The law of a path is a polynomial in q with score $(J - (N-1)q)/(pq)$, so $m'(q) = \mathbb{E}[X^2(J - (N-1)q)]/(pq) = \text{Cov}(J, X^2)/(pq)$. This is not zero because $R_N^2(q) > 0$. So m is strictly monotone near q . By the law of large numbers the mean of the X_i^2 tends to $m(q)$, so with probability tending to 1 it lies in the image of a small interval around q , and the solution in that

interval converges to q . The central limit theorem and the delta method give the limit law with variance

$$\frac{\text{Var } X^2}{m'(q)^2} = \frac{(pq)^2 \text{Var } X^2}{\text{Cov}(J, X^2)^2} = \frac{(pq)^2}{R_N^2(q) \text{Var } J} = \frac{pq}{(N-1)R_N^2(q)}.$$

Since $I_K(q) = e_N(q)(N-1)/(pq)$, the ratio of $1/I_K(q)$ to this variance is R_N^2/e_N , and it is at most 1 by (1). For the lower bound let C_1 be the constant in the expansion of Theorem 2.3, and put $C' = 4 \max(C, C_1)$ and $\epsilon = T^{-2} + T^4/N$. If $C'\epsilon \geq 1$ the claim is trivial, since $R_N^2/e_N \geq 0$. Otherwise $C_1\epsilon < \frac{1}{4}$, so the expansion gives $e_N \geq \frac{1}{2T} - \frac{C_1\epsilon}{T} \geq \frac{1}{4T}$. Then $1 - R_N^2/e_N = (e_N - R_N^2)/e_N \leq 4T \cdot C\epsilon/T = 4C\epsilon \leq C'\epsilon$. $\square$

A Proof of the Fisher information theorem

We keep the notation of Section 2. Let $\sigma_i \in \{-1, 1\}$ be the direction of step i . Then $X = \sigma_1 + \dots + \sigma_N$, and J counts the $i < N$ with $\sigma_{i+1} \neq \sigma_i$. Put $Z = \mathbb{E}[J \mid K]$, so $e_N = \text{Var } Z / \text{Var } J$ by Proposition 2.1.

Lemma A.1 (exact moments). *For $N \geq 1$ and $0 < q < 1$,*

$$\begin{aligned} \mathbb{E}X^2 &= \frac{Np}{q} - \frac{\rho(1 - \rho^N)}{2q^2}, \\ \text{Cov}(J, X^2) &= -\frac{p}{q^2} [Nq(1 + \rho^N) - p(1 - \rho^N)], \\ \mathbb{E}X^4 &= \frac{3N^2p^2}{q^2} - \frac{Np(1 + 10\rho + \rho^2)}{2q^3} + \frac{\rho(1 + 2\rho)(2 + \rho)(1 - \rho^N)}{2q^4} - \frac{3Np\rho^{N+1}}{q^3}, \end{aligned}$$

and $\text{Var } X^2 = 2N^2p^2V/q^2$, with V as in Theorem 2.3.

Proof. Let $m_N(\phi, \chi) = \mathbb{E}e^{\phi X + \chi J}$. With $A_{ab} = \mathbb{P}(\sigma_2 = b \mid \sigma_1 = a)e^{\phi b + \chi \mathbf{1}[a \neq b]}$ and $v_a = \frac{1}{2}e^{\phi a}$ we have $m_N = v^\top A^{N-1} \mathbf{1}$. So $\sum_{N \geq 1} m_N z^N = U/D$ as formal power series, where $U(z) = zv^\top \text{adj}(I - zA) \mathbf{1}$ and $D(z) = \det(I - zA)$ have degree at most 2 in z . For a partial derivative ∂ of order m in (ϕ, χ) , induction and the quotient rule give $\partial(U/D) = U_\partial/D^{m+1}$ with $\deg U_\partial \leq 2m + 2$, as each derivative multiplies the numerator by D or a derivative of D . At $\phi = \chi = 0$ the matrix A has trace $1 + \rho$ and determinant ρ , so $D(z) = (1 - z)(1 - \rho z)$. Let $q \neq \frac{1}{2}$, so its roots 1 and $1/\rho$ are distinct. By partial fractions U_∂/D^{m+1} is a constant plus a combination of $(1 - z)^{-j}$ and $(1 - \rho z)^{-j}$ with $j \leq m + 1$, whose coefficients of z^N are $\binom{N+j-1}{j-1}$ and $\binom{N+j-1}{j-1}\rho^N$. So for $N \geq 1$ the sequence ∂m_N at 0 lies in $\mathcal{S}_m = \text{span}\{N^i, N^i\rho^N : i \leq m\}$. A sequence in $\mathcal{S}_m$ satisfies the recurrence with characteristic polynomial $(t - 1)^{m+1}(t - \rho)^{m+1}$, so it is 0 if it vanishes at $N = 1, \dots, 2m + 2$.

The moments $\mathbb{E}X^2$, $\mathbb{E}[JX^2]$ and $\mathbb{E}X^4$ are derivatives of m_N of order at most 4 at 0, so they lie in $\mathcal{S}_4$. The claimed formulas for $\mathbb{E}X^2$, $\mathbb{E}X^4$ and $\mathbb{E}[JX^2] = \text{Cov}(J, X^2) + (N-1)q\mathbb{E}X^2$ lie in $\mathcal{S}_2$. So it is enough to check them at $N = 1, \dots, 10$. The script `paperC/verify_sticky_fisher.py` computes the three moments exactly for $N \leq 12$ by a dynamic program with q as a symbol, and every difference is 0. For fixed N both sides are rational in q with no pole at $\frac{1}{2}$, so the case $q = \frac{1}{2}$ follows. Finally, with R standing for ρ^N and ρR for ρ^{N+1} , the claim $\mathbb{E}X^4 - (\mathbb{E}X^2)^2 = 2N^2p^2V/q^2$ is an identity of rational functions of N , q and R , which the same script checks with sympy. $\square$

Proof of the closed form in Theorem 2.3. For $N \geq 2$, X^2 is N^2 on the constant paths and $(N-2)^2$ on the paths that switch only at the last step, so $\text{Var } X^2 > 0$ and $V > 0$ by Lemma A.1, which also gives $\text{Cov}(J, X^2) = -\frac{Np}{q}[1 + \rho^N - p(1 - \rho^N)/T]$, so $\text{Cov}(J, X^2)^2 / \text{Var } X^2$ is the squared bracket over $2V$. We divide by $\text{Var } J = (N-1)pq$. $\square$

Lemma A.2 (convexity). *Let f and g be convex and differentiable on $[t-\delta, t+\delta]$, with $0 \leq g-f \leq \eta$ and $g'' \leq \kappa$ there. Then $|f'(t) - g'(t)| \leq \eta/\delta + \kappa\delta/2$.*

Proof. By convexity and Taylor's theorem, $f'(t) \leq (f(t+\delta) - f(t))/\delta \leq (g(t+\delta) - g(t) + \eta)/\delta \leq g'(t) + \kappa\delta/2 + \eta/\delta$, and the lower bound is the same with $t - \delta$. $\square$

Lemma A.3 (a Bessel ratio). *For $t > 0$ let $R(t) = I_1(t)/I_0(t)$. Then $(\sqrt{1+t^2} - 1)/t \leq R(t) < 1$. In particular $0 < 1 - R(t) \leq 1/t$.*

Proof. Since $I'_0 = I_1$ and $I'_1(t) = I_0(t) - I_1(t)/t$, we have $R' = 1 - R/t - R^2$, and $R(t) = t/2 + O(t^3)$ at 0. At a first point t_0 with $R(t_0) = 1$ we would have $R'(t_0) \geq 0$, but the equation gives $-1/t_0$. So $R < 1$. Put $s = \sqrt{1+t^2}$ and $\ell = (s-1)/t$. Then $1 - \ell/t - \ell^2 = \ell/t = 1/(s+1)$ and $\ell' = 1/(s(s+1))$, so $\ell' \leq 1 - \ell/t - \ell^2$. For $d = R - \ell$ this gives $d' \geq -d/t - (R^2 - \ell^2) = -d(1/t + R + \ell)$. So Θd is nondecreasing, where $\Theta(t) = t \exp \int_0^t (R + \ell)$. As $\ell = t/2 + O(t^3)$, we have $d = O(t^3)$ and $\Theta d \rightarrow 0$ at 0. Hence $d \geq 0$. Finally $1 - \ell = (t + 1 - s)/t \leq 1/t$. $\square$

Lemma A.4 (moments of the endpoint). (a) *For each $m \geq 1$ there is C_m with $\mathbb{E}W^{2m} \leq C_m T^{-m}$ for all $0 < q < 1$ and $T \geq 1$.*

(b) *If $0 < q \leq \frac{1}{2}$, then $\mathbb{E}W^4 \leq 3p^2/T^2 + 9/(2T^4)$.*

Proof. (a) Add a step σ_{N+1} to the walk. Since $\mathbb{E}[\sigma_{i+1} | \sigma_{\leq i}] = \rho\sigma_i$, the $\xi_{i+1} = \sigma_{i+1} - \rho\sigma_i$ are martingale differences. Summing $\sigma_{i+1} - \sigma_i = -2q\sigma_i + \xi_{i+1}$ over $i \leq N$ gives $X = (\sigma_1 - \sigma_{N+1} + \Xi)/(2q)$ with $\Xi = \sum_{i=2}^{N+1} \xi_i$. Also ξ_{i+1}^2 is $4q^2$ if step $i+1$ repeats step i and $4p^2 \leq 4$ otherwise. So $\sum \xi_i^2 \leq 4q^2N + 4J'$, where $J' \sim \text{Bin}(N, q)$ counts the switches among the $N+1$ steps. Burkholder's inequality [3, Ch. 2] gives $\mathbb{E}\Xi^{2m} \leq c_m \mathbb{E}(4q^2N + 4J')^m \leq c_m 8^m ((q^2N)^m + \mathbb{E}J'^m)$, and $q^2N \leq T$. The falling factorial moments of J' are at most T^j , and J'^m is a combination of the falling factorials of order $j \leq m$ with nonnegative coefficients. So $\mathbb{E}J'^m \leq b_m T^m$ for $T \geq 1$. As $|\sigma_1 - \sigma_{N+1}| \leq 2$, we get $\mathbb{E}X^{2m} \leq (2q)^{-2m} 2^{2m-1} (2^{2m} + \mathbb{E}\Xi^{2m}) \leq C_m q^{-2m} T^m$, and $\mathbb{E}W^{2m} = N^{-2m} \mathbb{E}X^{2m} \leq C_m T^{-m}$.

(b) We divide the formula for $\mathbb{E}X^4$ by N^4 . As $0 \leq \rho < 1$, the two terms of order T^{-3} are at most 0 and the one of order T^{-4} is at most $9/(2T^4)$. $\square$

Proof of the bound in Theorem 2.3. Let $\mathcal{P}$ be the orthogonal projection in L^2 onto the span of 1 and X^2 . Since X^2 is a function of K , $\text{Cov}(Z, X^2) = \text{Cov}(J, X^2)$. So $\text{Var } Z = \text{Cov}(J, X^2)^2 / \text{Var } X^2 + \mathbb{E}(Z - \mathcal{P}Z)^2$, and $e_N - R_N^2 = \mathbb{E}(Z - \mathcal{P}Z)^2 / \text{Var } J \geq 0$. Also $\mathbb{E}(Z - \mathcal{P}Z)^2 \leq \mathbb{E}(Z - G)^2$ for $G = y(1 - W^2/2) + \frac{1}{2}$. From now on $0 < q \leq \frac{1}{2}$ and $1 \leq T \leq N^{1/4}$, and C is an absolute constant that may change at each use. Then $p \geq \frac{1}{2}$, $T \leq y \leq 2T$ and $\text{Var } J = (N-1)pq \geq T/4$. It suffices to show $\mathbb{E}(Z - G)^2 \leq C(T^{-2} + T^4/N)$.

For $0 < k < N$ put $s_k = \sqrt{k(N-k)}$, $c_k = N/(2s_k)$ and $w_k = 2s_k u$. On $K = k$ we have $s_k^2 = N^2(1 - W^2)/4$, so $c_k = (1 - W^2)^{-1/2} \geq 1$, $w_k = y\sqrt{1 - W^2}$ and $c_k w_k = y$. Let $\mathcal{I}_k(z) = I_0(2z) + c_k I_1(2z)$ and $\tilde{S}_k(v) = 2v \mathcal{I}_k(s_k v)$, which approximates $S_{k, N-k}(v)$. Here I_0 and I_1 are the modified Bessel functions. By the uniform Bessel estimate for every bin in [2] (the theorem “uniform estimate for every bin” there, with $a = k$ and $b = N - k$),

$$0 \leq 1 - \frac{S_{k, N-k}(v)}{\tilde{S}_k(v)} \leq \frac{Nv^2}{2} + v \quad \text{for all } v > 0. \quad (2)$$

Let $\Lambda_k(t) = \log S_{k, N-k}(e^t)$ and $\beta_k(t) = \log \tilde{S}_k(e^t)$. By Proposition 2.1, $Z = \Lambda'_K(\log u)$ on $0 < K < N$.

Step 1. Both Λ_k and β_k are convex, as logarithms of series in e^t with nonnegative coefficients. Let $|t - \log u| \leq 1$. Then $v = e^t \leq eu$ and $y \leq 2N^{1/4}$ give $Nv^2/2 + v \leq (e^2y^2/2 + ey)/N \leq (2e^2 + 2e)/\sqrt{N} \leq \frac{1}{2}$ for $N \geq 1700$. As $-\log(1 - \varepsilon) \leq 2\varepsilon$ for $0 \leq \varepsilon \leq \frac{1}{2}$, (2) gives $0 \leq \beta_k - \Lambda_k \leq \eta$ there, with $\eta = (e^2y^2 + 2ey)/N$. Next, $\beta_k(t) - t - \log 2$ is the logarithm of $\mathcal{I}_k(s_k e^t) = \sum_n a_n e^{nt}$ with $a_n \geq 0$. So β_k'' is the variance of n under the weights $a_n e^{nt}$, at most the mean of n^2 . As $(z \frac{d}{dz})^2$ maps $I_0(2z)$ to $4z^2 I_0(2z)$ and $I_1(2z)$ to $(4z^2 + 1)I_1(2z)$, this mean is at most $4z^2 + 1 \leq e^2y^2 + 1 = \kappa$, since $z = s_k e^t \leq ey/2$. For $N \geq 4$, $\delta = \sqrt{2\eta/\kappa} \leq 1$ and Lemma A.2 gives

$$|\Lambda'_k(\log u) - \beta'_k(\log u)| \leq \sqrt{2\eta\kappa} \leq \frac{2e^2y^2 + 2ey + 1}{\sqrt{2N}} \leq \frac{11(1+y)^2}{\sqrt{N}}. \quad (3)$$

Step 2. Write $z = s_k e^t$, $w = 2z$ and $R = R(w)$. By the formulas for I'_0 and I'_1 ,

$$\beta'_k(t) = 1 + \frac{z\mathcal{I}'_k(z)}{\mathcal{I}_k(z)} = 1 + \frac{wR + c_k w - c_k R}{1 + c_k R} = w + \frac{1 + w(c_k - 1)(1 - R)}{1 + c_k R}.$$

At $t = \log u$ we have $w = w_k$, so $\beta'_k(\log u) = w_k + B_k$ with B_k the last fraction. By Lemma A.3, $0 < R < 1$ and $w(1 - R) \leq 1$. So the numerator of B_k lies in $[1, c_k]$ and is at most $1 + wc_k = 1 + y$. Hence $0 < B_k \leq 1 + y$. Subtracting $\frac{1}{2}$ from $1/(1 + c_k R) \leq B_k \leq c_k/(1 + c_k R)$ gives

$$-\frac{c_k - 1}{2} \leq B_k - \frac{1}{2} \leq \frac{(c_k - 1) + c_k(1 - R)}{2(1 + c_k R)} \leq \frac{c_k - 1}{2} + \frac{c_k^2}{2y},$$

by $c_k(1 - R) \leq c_k/w_k = c_k^2/y$. If $W^2 \leq \frac{1}{2}$ on $K = k$, then $c_k^2 \leq 2$ and $c_k - 1 \leq W^2$, since $(1 - v)^{-1/2} - 1$ is convex, vanishes at 0 and is below $\frac{1}{2}$ at $v = \frac{1}{2}$. So $|B_k - \frac{1}{2}| \leq W^2/2 + 1/y$ there.

Step 3. Let $\psi(a) = 1 - a^2/2 - \sqrt{1 - a^2}$ for $|a| \leq 1$. The series of $\sqrt{1 - v}$ is $1 - v/2 - \sum_{j \geq 2} b_j v^j$ with $b_j \geq 0$ and $\sum_j b_j = \frac{1}{2}$, so $0 \leq \psi(a) \leq a^4/2$. On $K = k$, $w_k = y(1 - W^2/2) - y\psi(W)$. So on $0 < K < N$ we have $Z - G = (Z - \beta'_K(\log u)) + (B_K - \frac{1}{2}) - y\psi(W)$, and on the atoms $K \in \{0, N\}$ we have $Z = 0$, $W^2 = 1$ and $Z - G = -(y + 1)/2$.

Step 4. By Lemma A.4 and Markov's inequality, $\mathbb{E}W^4 \leq 8/T^2$, $\mathbb{E}W^8 \leq CT^{-4}$ and $\mathbb{P}(W^2 > \frac{1}{2}) \leq 16\mathbb{E}W^8 \leq CT^{-4}$. Also $\mathbb{P}(K \in \{0, N\}) = p^{N-1} \leq e^{-q(N-1)} \leq e^{1/2-T}$.

Step 5. By Step 3 and Minkowski's inequality, $\|Z - G\|_2$ is at most the sum of the L^2 norms of the three terms of Step 3 on $0 < K < N$ and of $\frac{y+1}{2}$ on the atoms. By (3) the first is at most $99T^2/\sqrt{N}$, as $(1 + y)^2 \leq 9T^2$. For the second we use Step 2 where $W^2 \leq \frac{1}{2}$ and $|B_K - \frac{1}{2}| \leq 1 + y$ elsewhere, so its square is at most $\mathbb{E}W^4/2 + 2/y^2 + C(1 + y)^2T^{-4} \leq C/T^2$. The third is at most $\frac{y}{2}(\mathbb{E}W^8)^{1/2} \leq C/T$, and the last is $\frac{y+1}{2}\mathbb{P}(K \in \{0, N\})^{1/2} \leq CT e^{-T/2} \leq C/T$. So $\mathbb{E}(Z - G)^2 \leq C(T^4/N + T^{-2})$, and dividing by $\text{Var } J \geq T/4$ gives the bound. Only Step 1 used $N \geq 1700$, so $N_0 = 1700$ works. $\square$

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